

Answer key - practice sets 1 to 5 (Graph Neural Networks, Fall 2026)

모든 숫자는 numpy 로 계산한 값이다. 서술형 a 는 개념 문제라 답을 적지 않았다. 사이트 시험 대비 화면의 풀이에서 영어 문장 틀을 가져다 쓰면 된다.

연습 1회

p.1

b) $P(Y1) = 1, P(Y2) = 0.6667, P(Y3) = 0.25, P(Y4) = 0.5, P(Y5) = 0$

c) $P(Y1) = 1, P(Y2) = 0.5833, P(Y3) = 0.3333, P(Y4) = 0.5556, P(Y5) = 0$

p.2

b) $q(A) = 0.6237, q(B) = 0.0672, q(C) = 0.0672, q(D) = 0.2177, q(E) = 0.0242$

c) $h1(A) = 0$

p.3

b) $N(E) = 1, p(B|E) = 0.3333, p(C|E) = 0.3333, p(D|E) = 0.3333$

c) $E = 1.3333, h1(E) = 1.3333$

p.4

b) $Atilde = [0, 0.5, 0.5, 0]; [0.5, 0, 0, 0.5]; [0.5, 0, 0, 0.5]; [0, 0.5, 0.5, 0]$

$Z1 = [0.4, 0.2, 0, 0.4]$

c) $Z2 = [0.08, 0.52, 0.32, 0.08]$

p.5

b) $H1 = [0, 3, 3, 0], H2 = [3, 0, 0, 3]$

c) $Z = [2.2, -0.5, 2.3, 0.8]$

p.6

b) $A0H = [1, -3, 3, -1], A1H = [-3, 1, -2, 0], A2H = [1, -1.6667, 0.5, -0.5]$

c) $H1 =$ 세 조각을 이어 붙임 $[1, 0, 3, 0] \parallel [0, 1, 0, 0] \parallel [1, 0, 0.5, 0]$

$\Delta = [-1, 1, -0.5, 0]$

p.7

b) 평균 = $[3.25, 0.5]$, 중심 뺀 것 = $[-1.25, -2.5]; [0.75, -1.5]; [0.75, 2.5]; [-0.25, 1.5]$

c) 분모 = 2.222, $x = [-0.5625, -1.1251]; [0.3375, -0.6751]; [0.3375, 1.1251]; [-0.1125, 0.6751]$

p.8

b) 기대 선 수 = 4, $H1 = [0.6667, 2.6667, 0.6667, 1]$

c) $H1 = [0, 2.5, 2, 1]$

p.9

b) $\pi(B \rightarrow A) = 0.5, \pi(B \rightarrow C) = 2, \pi(B \rightarrow E) = 2$

c) $P(A) = 0.1111, P(C) = 0.4444, P(E) = 0.4444$

p.10

b) $p2(A|C) = 0.1412, p2(B|C) = 0.1806, p2(C|C) = 0.3819, p2(D|C) = 0.2963$

p.11

b) $Atilde = [0, 0, 0.5, 0.5, 0]; [0, 0, 0.5, 0.5, 0]; [0.25, 0.25, 0, 0.25, 0.25]; [0.3333, 0.3333, 0.3333, 0, 0]; [0, 0, 1, 0, 0]$

c) $H1 = [2.5, 2.5, 0.75, 1.3333, 2]$

p.12

b) $h(A) = [3, 0.5], h(B) = [4, 0.5], h(C) = [0, 3.25], h(D) = [2, 2], h(E) = [4, 0]$

c) $hG = [2.6, 1.25]$

p.13

b) $h(A) = [2, 0], h(B) = [0, 0], h(C) = [0, 1], h(D) = [4, 2], h(E) = [3, 2.5], h(F) = [1, 3.5]$

p.14

b) $h(A) = [0, 2.5]$, $h(B) = [4, 3]$, $h(C) = [2, 0]$, $h(D) = [3, 1.5]$

p.15

b) $\text{Atilde} = [0, 0.3333, 0.3333, 0.3333, 0]; [0.3333, 0, 0, 0.3333, 0.3333]; [0.5, 0, 0, 0.5, 0]; [0.3333, 0.3333, 0.3333, 0, 0]; [0, 1, 0, 0, 0]$

$H1 = [0.3333, 0.3333, 0, 0, 2]$

c) $H2 = [0.3333, 0.7778, 0.1667, 0.2222, 2]$

p.16

b) $\text{Atilde} = [0, 0, 0.3333, 0.3333, 0.3333]; [0, 0, 0, 0, 1]; [0.5, 0, 0, 0.5, 0]; [0.3333, 0, 0.3333, 0, 0.3333]; [0.3333, 0.3333, 0, 0.3333, 0]$

c) $H1 = [0.1667, 0, 0.2, 0.3667, 0]$

p.17

b) $H1 = [1, 5, 4, 2.6667]$

$H2 = [4.3333, 8.3333, 6.8889, 6]$

c) plain GCN $H1 = [2, 2, 1, 1.6667]$ (잔차가 없으면 자기 값과 음수가 사라진다)

연습 2회

p.18

b) $P(Y1) = 1$, $P(Y2) = 0.6667$, $P(Y3) = 0.75$, $P(Y4) = 0.75$, $P(Y5) = 0$

c) $P(Y1) = 1$, $P(Y2) = 0.8333$, $P(Y3) = 0.8333$, $P(Y4) = 0.8333$, $P(Y5) = 0$

p.19

b) $q(A) = 0.4889$, $q(B) = 0.1889$, $q(C) = 0.0444$, $q(D) = 0.1889$, $q(E) = 0.0889$

c) $h1(A) = 0.1667$

p.20

b) $N(C) = 1$, $p(A|C) = 0.3333$, $p(D|C) = 0.3333$, $p(E|C) = 0.3333$

c) $E = 2.3333$, $h1(C) = 2.3333$

p.21

b) $\text{Atilde} = [0, 0.5, 0.5, 0]; [0.5, 0, 0.5, 0]; [0.3333, 0.3333, 0, 0.3333]; [0, 0, 1, 0]$

$Z1 = [0, 0, 0.1667, 0.5]$

c) $Z2 = [0.0417, 0.0417, 0.0833, 0.5833]$

p.22

b) $H1 = [-1.5, -2, 0.5, 0.6667]$, $H2 = [0.5833, 0.6667, -0.4167, -1]$

c) $Z = [1.5333, -0.7333, -0.4667, -1]$

p.23

b) $A0H = [2, 3, 4, -1]$, $A1H = [1, 0.5, -1, 3]$, $A2H = [1.75, 2, 3, 0.1667]$

c) $H1 = \text{세 조각을 이어 붙임 } [2, 3, 4, 0] \parallel [1, 0.5, 0, 3] \parallel [1.75, 2, 3, 0.1667]$

$\text{Delta} = [-0.75, -1.5, -3, 2.8333]$

p.24

b) 평균 = $[0, 1.5]$, 중심 뺀 것 = $[0, 2.5]; [1, 1.5]; [1, -3.5]; [-2, -0.5]$

c) 분모 = 2.5981, $x = [0, 0.9623]; [0.3849, 0.5774]; [0.3849, -1.3472]; [-0.7698, -0.1925]$

p.25

b) 기대 선 수 = 4, $H1 = [0.6667, 1.3333, 0, 0]$

c) $H1 = [0, 1.3333, 0, 0]$

p.26

b) $\pi(B \rightarrow A) = 0.5$, $\pi(B \rightarrow C) = 2$, $\pi(B \rightarrow E) = 1$

c) $P(A) = 0.1429$, $P(C) = 0.5714$, $P(E) = 0.2857$

p.27

b) $p_2(A|C) = 0.2963$, $p_2(B|C) = 0.1412$, $p_2(C|C) = 0.1806$, $p_2(D|C) = 0.3819$

p.28

b) $\text{Atilde} = [0, 0.5, 0.5, 0, 0]; [0.5, 0, 0, 0, 0.5]; [0.3333, 0, 0, 0.3333, 0.3333]; [0, 0, 0.5, 0, 0.5]; [0, 0.3333, 0.3333, 0.3333, 0]$

c) $H1 = [0, 1.5, 0.3333, 3, 0]$

p.29

b) $h(A) = [0, 0]$, $h(B) = [3, 0]$, $h(C) = [0, 0.5]$, $h(D) = [0, 0]$, $h(E) = [0, 0.5]$

c) $hG = [0.6, 0.2]$

p.30

b) $h(A) = [3, 4]$, $h(B) = [4, 3]$, $h(C) = [0, 0]$, $h(D) = [2, 0]$, $h(E) = [1, 0]$, $h(F) = [0, 1]$

p.31

b) $h(A) = [4, 1.5]$, $h(B) = [4, 3]$, $h(C) = [2, 4]$, $h(D) = [0, 4]$

p.32

b) $\text{Atilde} = [0, 0.3333, 0, 0.3333, 0.3333]; [0.5, 0, 0, 0.5, 0]; [0, 0, 0, 0, 1]; [0.3333, 0.3333, 0, 0, 0.3333]; [0.3333, 0, 0.3333, 0.3333, 0]$

$H1 = [0, 1, 0, 0.6667, 0]$

c) $H2 = [0.5556, 1, 0, 0.6667, 0.2222]$

p.33

b) $\text{Atilde} = [0, 0, 0, 0.5, 0.5]; [0, 0, 0.5, 0, 0.5]; [0, 0.5, 0, 0.5, 0]; [0.3333, 0, 0.3333, 0, 0.3333]; [0.3333, 0.3333, 0, 0.3333, 0]$

c) $H1 = [0, 0, 0, 0, 0]$

p.34

b) $H1 = [-1.5, 0.6667, 3.5, 1.3333]$

$H2 = [-0.5, 1.7778, 4.5, 2.2222]$

c) plain GCN $H1 = [0.5, 0.6667, 0.5, 0.3333]$ (잔차가 없으면 자기 값과 음수가 사라진다)

연습 3회

p.35

b) $P(Y1) = 1$, $P(Y2) = 0.5$, $P(Y3) = 0.5$, $P(Y4) = 0.75$, $P(Y5) = 0$

c) $P(Y1) = 1$, $P(Y2) = 0.5833$, $P(Y3) = 0.5$, $P(Y4) = 0.75$, $P(Y5) = 0$

p.36

b) $q(A) = 0.0444$, $q(B) = 0.4889$, $q(C) = 0.0889$, $q(D) = 0.1889$, $q(E) = 0.1889$

c) $h1(A) = 0$

p.37

b) $N(C) = 1$, $p(A|C) = 0.3333$, $p(D|C) = 0.3333$, $p(E|C) = 0.3333$

c) $E = -0.6667$, $h1(C) = 0$

p.38

b) $\text{Atilde} = [0, 0, 0.5, 0.5]; [0, 0, 0.5, 0.5]; [0.5, 0.5, 0, 0]; [0.5, 0.5, 0, 0]$

$Z1 = [0.25, 0.25, 0, 0.5]$

c) $Z2 = [0.125, 0.125, 0.125, 0.625]$

p.39

b) $H1 = [1, 1.5, 1, 1.5]$, $H2 = [1.5, 1, 1.5, 1]$

c) $Z = [1.9, -0.9, -0.6, 2.1]$

p.40

b) $A0H = [2, -2, 1, -1]$, $A1H = [0, -1, 0.5, 0.3333]$, $A2H = [0.4167, 0.3333, 0.1667, -0.1667]$

c) $H1 = \text{세 조각을 이어 붙임 } [2, 0, 1, 0] \mid [0, 0, 0.5, 0.3333] \mid [0.4167, 0.3333, 0.1667, 0]$

$\Delta = [-0.4167, -0.3333, 0.3333, 0.3333]$

p.41

- b) 평균 = [1.25, 1.5], 중심 뺀 것 = [2.75, -1.5]; [-0.25, 1.5]; [-0.25, -2.5]; [-2.25, 2.5]
c) 분모 = 2.7272, $x = [1.0084, -0.55]; [-0.0917, 0.55]; [-0.0917, -0.9167]; [-0.825, 0.9167]$

p.42

- b) 기대 선 수 = 4, $H1 = [1.3333, 2.3333, 0, 0.3333]$
c) $H1 = [0, 2, 0, 2]$

p.43

- b) $\pi(B \rightarrow A) = 0.5, \pi(B \rightarrow C) = 2, \pi(B \rightarrow D) = 1$
c) $P(A) = 0.1429, P(C) = 0.5714, P(D) = 0.2857$

p.44

- b) $p2(A|B) = 0.1806, p2(B|B) = 0.2963, p2(C|B) = 0.3819, p2(D|B) = 0.1412$

p.45

- b) $\tilde{A} = [0, 0, 0.5, 0, 0.5]; [0, 0, 0.3333, 0.3333, 0.3333]; [0.3333, 0.3333, 0, 0, 0.3333]; [0, 1, 0, 0, 0]; [0.3333, 0.3333, 0.3333, 0, 0]$
c) $H1 = [1.5, 2.3333, 1, 0, 0]$

p.46

- b) $h(A) = [0, 1], h(B) = [3, 0.6667], h(C) = [4, 0.5], h(D) = [0, 2.3333], h(E) = [2, 0]$
c) $hG = [1.8, 0.9]$

p.47

- b) $h(A) = [0, 0], h(B) = [3, 1], h(C) = [1, 3], h(D) = [2, 4], h(E) = [4, 2], h(F) = [2, 4]$

p.48

- b) $h(A) = [1, 0], h(B) = [0, 0], h(C) = [0, 0], h(D) = [2, 0]$

p.49

- b) $\tilde{A} = [0, 0.25, 0.25, 0.25, 0.25]; [0.3333, 0, 0, 0.3333, 0.3333]; [1, 0, 0, 0, 0]; [0.5, 0.5, 0, 0, 0]; [0.5, 0.5, 0, 0, 0]$
 $H1 = [0, 1.3333, 4, 0.5, 0.5]$
c) $H2 = [1.5833, 1.3333, 4, 0.6667, 0.6667]$

p.50

- b) $\tilde{A} = [0, 0.5, 0.5, 0, 0]; [0.3333, 0, 0, 0.3333, 0.3333]; [0.3333, 0, 0, 0.3333, 0.3333]; [0, 0.5, 0.5, 0, 0]; [0, 0.5, 0.5, 0, 0]$
c) $H1 = [1.08, 0, 0, 1.44, 1.32]$

p.51

- b) $H1 = [-1.5, 1, 2, 1.5]$
 $H2 = [0, 1.6667, 2.3333, 3]$
c) plain GCN $H1 = [1.5, 0, 0, 1.5]$ (잔차가 없으면 자기 값과 음수가 사라진다)

연습 4회

p.52

- b) $P(Y1) = 1, P(Y2) = 0.25, P(Y3) = 0.3333, P(Y4) = 0.75, P(Y5) = 0$
c) $P(Y1) = 1, P(Y2) = 0.1667, P(Y3) = 0.3333, P(Y4) = 0.6667, P(Y5) = 0$

p.53

- b) $q(A) = 0.2821, q(B) = 0.1667, q(C) = 0.1667, q(D) = 0.2821, q(E) = 0.1026$
c) $h1(A) = 0.3333$

p.54

- b) $N(B) = 1, p(A|B) = 0.3333, p(C|B) = 0.3333, p(D|B) = 0.3333$
c) $E = 2, h1(B) = 2$

p.55

- b) $\text{Atilde} = [0, 0.3333, 0.3333, 0.3333]; [1, 0, 0, 0]; [0.5, 0, 0, 0.5]; [0.5, 0, 0.5, 0]$
 $Z1 = [0.2, 0.8, 0.4, 0.4]$
c) $Z2 = [0.6267, 0.16, 0.24, 0.24]$

p.56

- b) $H1 = [0.5, 0, 1, 3], H2 = [0.5, 1.5, 0.25, 0]$
c) $Z = [-0.4, 1.2, -0.65, 2.7]$

p.57

- b) $A0H = [-2, -2, -2, 3], A1H = [-2, 0.5, 0.5, -2], A2H = [0.5, -2, -2, 0.5]$
c) $H1 = \text{세 조각을 이어 붙임 } [0, 0, 0, 3] \mid [0, 0.5, 0.5, 0] \mid [0.5, 0, 0, 0.5]$
 $\text{Delta} = [-0.5, 0.5, 0.5, -0.5]$

p.58

- b) 평균 = $[0.25, 0]$, 중심 뺀 것 = $[-1.25, 0]; [-0.25, 0]; [3.75, 0]; [-2.25, 0]$
c) 분모 = 2.2776, $x = [-0.5488, 0]; [-0.1098, 0]; [1.6465, 0]; [-0.9879, 0]$

p.59

- b) 기대 선 수 = 4, $H1 = [2, 1.6667, 0.6667, 0.6667]$
c) $H1 = [0, 1.6667, 1.5, 1.5]$

p.60

- b) $\text{pi}(B \rightarrow A) = 0.5, \text{pi}(B \rightarrow C) = 1, \text{pi}(B \rightarrow D) = 2$
c) $P(A) = 0.1429, P(C) = 0.2857, P(D) = 0.5714$

p.61

- b) $p2(A|B) = 0.1806, p2(B|B) = 0.1412, p2(C|B) = 0.2963, p2(D|B) = 0.3819$

p.62

- b) $\text{Atilde} = [0, 0.25, 0.25, 0.25, 0.25]; [0.5, 0, 0, 0, 0.5]; [0.5, 0, 0, 0.5, 0]; [0.5, 0, 0.5, 0, 0]; [0.5, 0.5, 0, 0, 0]$
c) $H1 = [0, 0, 0.5, 0, 0]$

p.63

- b) $h(A) = [0, 0.3333], h(B) = [0, 0], h(C) = [0, 0], h(D) = [0, 0.3333], h(E) = [4, 0]$
c) $hG = [0.8, 0.1333]$

p.64

- b) $h(A) = [0, 0], h(B) = [2, 0], h(C) = [0, 2], h(D) = [0, 2], h(E) = [2, 0], h(F) = [0, 0]$

p.65

- b) $h(A) = [4, 2], h(B) = [0, 2], h(C) = [1, 2], h(D) = [2, 1]$

p.66

- b) $\text{Atilde} = [0, 0.3333, 0, 0.3333, 0.3333]; [0.3333, 0, 0.3333, 0.3333, 0]; [0, 0.5, 0, 0, 0.5]; [0.5, 0.5, 0, 0, 0]; [0.5, 0, 0.5, 0, 0]$
 $H1 = [0.3333, 0, 1.5, 1, 1]$
c) $H2 = [0.6667, 0.9444, 1.5, 1, 1]$

p.67

- b) $\text{Atilde} = [0, 0.5, 0, 0.5, 0]; [0.3333, 0, 0.3333, 0.3333, 0]; [0, 0.5, 0, 0, 0.5]; [0.3333, 0.3333, 0, 0, 0.3333]; [0, 0, 0.5, 0.5, 0]$
c) $H1 = [1.625, 1.3333, 0, 1.25, 0]$

p.68

- b) $H1 = [4, -1.3333, 0, -1.3333]$
 $H2 = [4, -0.4444, 0, -0.4444]$
c) plain GCN $H1 = [0, 0.6667, 0, 0.6667]$ (잔차가 없으면 자기 값과 음수가 사라진다)

연습 5회

p.69

b) $P(Y1) = 1, P(Y2) = 0.6667, P(Y3) = 0.25, P(Y4) = 0.75, P(Y5) = 0$

c) $P(Y1) = 1, P(Y2) = 0.6667, P(Y3) = 0.3333, P(Y4) = 0.8333, P(Y5) = 0$

p.70

b) $q(A) = 0.0889, q(B) = 0.0444, q(C) = 0.1889, q(D) = 0.1889, q(E) = 0.4889$

c) $h1(A) = 0$

p.71

b) $N(E) = 1, p(A|E) = 0.5, p(D|E) = 0.5$

c) $E = 3, h1(E) = 3$

p.72

b) $Atilde = [0, 0.5, 0.5, 0]; [0.5, 0, 0, 0.5]; [0.5, 0, 0, 0.5]; [0, 0.5, 0.5, 0]$

$Z1 = [0.4, 0.2, 0, 0.4]$

c) $Z2 = [0.08, 0.52, 0.32, 0.08]$

p.73

b) $H1 = [-0.5, -1.5, 0.6667, -2], H2 = [-0.4167, 0.0833, -1.3333, 0.6667]$

c) $Z = [-0.675, 0.375, -0.8667, 0.8]$

p.74

b) $A0H = [3, -3, 1, 0], A1H = [-1, 1.5, 1.5, -1], A2H = [1.5, -1, -1, 1.5]$

c) $H1 = \text{세 조각을 이어 붙임 } [3, 0, 1, 0] \mid [0, 1.5, 1.5, 0] \mid [1.5, 0, 0, 1.5]$

$\Delta = [-1.5, 1.5, 1.5, -1.5]$

p.75

b) 평균 = $[-0.75, 1.75], \text{중심 뺀 것} = [-1.25, 2.25]; [1.75, -2.75]; [-0.25, -0.75]; [-0.25, 1.25]$

c) 분모 = 2.2079, $x = [-0.5661, 1.019]; [0.7926, -1.2455]; [-0.1132, -0.3397]; [-0.1132, 0.5661]$

p.76

b) 기대 선 수 = 4, $H1 = [1, 0, 0, 0.3333]$

c) $H1 = [1.5, 0, 0, 2]$

p.77

b) $\pi(B \rightarrow A) = 0.5, \pi(B \rightarrow C) = 1, \pi(B \rightarrow E) = 2$

c) $P(A) = 0.1429, P(C) = 0.2857, P(E) = 0.5714$

p.78

b) $p2(A|B) = 0.2963, p2(B|B) = 0.3819, p2(C|B) = 0.1412, p2(D|B) = 0.1806$

p.79

b) $Atilde = [0, 0.5, 0.5, 0, 0]; [0.3333, 0, 0, 0.3333, 0.3333]; [0.3333, 0, 0, 0.3333, 0.3333]; [0, 0.5, 0.5, 0, 0]; [0, 0.5, 0.5, 0, 0]$

c) $H1 = [2, 0, 0, 2, 2]$

p.80

b) $h(A) = [3, 0.3333], h(B) = [4, 1], h(C) = [0, 2.5], h(D) = [0, 3.5], h(E) = [2, 2]$

c) $hG = [1.8, 1.8667]$

p.81

b) $h(A) = [0, 4], h(B) = [4, 0], h(C) = [4, 0], h(D) = [2, 4], h(E) = [0, 4], h(F) = [4, 1]$

p.82

b) $h(A) = [3, 1], h(B) = [1, 3], h(C) = [3, 1], h(D) = [3, 1]$

p.83

b) $\tilde{A} = [0, 0.3333, 0.3333, 0, 0.3333]; [0.5, 0, 0, 0, 0.5]; [0.3333, 0, 0, 0.3333, 0.3333]; [0, 0, 1, 0, 0]; [0.3333, 0.3333, 0.3333, 0, 0]$

$H1 = [0.6667, 0, 0.6667, 4, 1.3333]$

c) $H2 = [0.6667, 1, 2, 4, 1.3333]$

p.84

b) $\tilde{A} = [0, 0, 0.5, 0.5, 0]; [0, 0, 0.5, 0, 0.5]; [0.3333, 0.3333, 0, 0.3333, 0]; [0.3333, 0, 0.3333, 0, 0.3333]; [0, 0.5, 0, 0.5, 0]$

c) $H1 = [1.35, 0, 1.1, 1.5, 0]$

p.85

b) $H1 = [-2, 1, 2, 0]$

$H2 = [-1, 1, 2, 0]$

c) plain GCN $H1 = [1, 0, 0, 0]$ (잔차가 없으면 자기 값과 음수가 사라진다)